

AV NETWORK DESIGN REPORT

V25 Default demo design

October 1, 2026

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NOT VALIDATED

Estimate for planning only. Send this design to the Pro AV Design team for validation before ordering.

82

DEVICES

7

SWITCHES

1,326 W

POE LOAD

2.4 kW

EST. POWER

9.0 U

RACK SPACE

Networking engineered for AV over IP

NETGEAR AV switches, routers and WiFi access points are built specifically for networked media and AV, with out-of-the-box performance and easy setup for even the most complex installation. NETGEAR works with AV manufacturers and integration partners around the world, and supports the partners and associations that make the Pro AV and broadcast industries work.

- **Optimized for AV over IP**

Low-latency switching for audio, video and control, with PoE+ (30 W) and Ultra PoE++ (90 W) per port options.

- **Certified AV profiles**

NETGEAR AV OS with configuration profiles certified by leading AV manufacturers (Audinate, Crestron, Q-SYS, SMPTE ST 2110, NDI).

- **Multicast that just works**

NETGEAR IGMP Plus stops multicast flooding automatically, and Auto-LAG and Auto-Trunk connect switches without manual setup.

- **Free central management**

The NETGEAR Engage Controller manages switches, routers and WiFi 7 access points from one screen, and can set up the network before the hardware arrives.

- **Scales with the project**

From a single rack to redundant, active-active cores with 10G, 25G and 100G uplinks.

- **Lower total cost**

Energy-efficient hardware, a lifetime warranty on managed switches and less need for specialist IT resources.

IN THIS DESIGN

This design uses M4250 and M4350 switches, WBE758 WiFi 7 access points and the PR460X Pro Router, all managed for free from one screen in the NETGEAR Engage Controller with NETGEAR AV OS.

BUILT FOR EVERY SPACE

Broadcast and live events · Corporate and commercial AV · Higher education · Medical · Unified communications · High-end residential

HELP FROM DESIGN TO DEPLOYMENT

NETGEAR has a dedicated Pro AV and broadcast network design team, pre-sales support and technical assistance. Free on-demand training and certification at academy.netgear.com covers networking basics through in-depth troubleshooting, and the top courses are AVIXA-accredited for CTS renewal units.

Support and services

Every NETGEAR managed switch includes a lifetime hardware warranty and complimentary support to get you started. Premium Support tiers add guaranteed response times, longer support hours and faster hardware replacement, so you can match support to how critical the system is.

	Warranty (included)	Sprint	Overdrive	Fastlane
Complimentary setup support	✓	✓	✓	✓
AI-driven chat support	✓	✓	✓	✓
Direct access to support engineers		✓	✓	✓
Support hours		24/5	24/7	24/7
Guaranteed response time (P1)		8 hours	2 hours	Custom
Priority case handling			✓	✓
Escalation to senior engineers			Level 2	Level 3
Hardware replacement	Fast RMA	Advance replacement	Next business day	Custom
Scheduled support calls			✓	✓
Operational reports			Quarterly	Custom
Designated or dedicated engineer				✓
Custom workflows and SLAs				✓

Managed and enterprise products include 90 days of complimentary phone, chat and online support. Fully managed switches include 3 years of Sprint Support. Sprint and Overdrive response times are engagement commitments for Priority 1 cases.

PROFESSIONAL SERVICES

Deployment services	Design validation, hands-on configuration and readiness testing, so the network is production-ready from day one. Remote, onsite or hybrid.
Health checks and optimization	Proactive network audits that find issues before they become outages, with a clear remediation plan.
Custom training and enablement	Training based on your deployed system: architecture walkthroughs, feature deep dives, best practices and troubleshooting.
Interoperability, staging and migration	Multi-vendor compatibility checks, staged configurations and low-risk migrations to NETGEAR.
Event services	Pre-event validation and real-time onsite or remote engineering support for live broadcasts, sporting events and other high-stakes shows.
Fastlane professional engagement	A high-touch engagement with senior engineering access for organizations that need deeper collaboration.
Part number	Service
PSP1104-10000S	On-site engineering assistance, full day
PRC0001-10000S	New remote configuration, full day

Pricing depends on project scope and the level of support, and includes travel and expenses. Ask your NETGEAR contact or distributor for a quote.

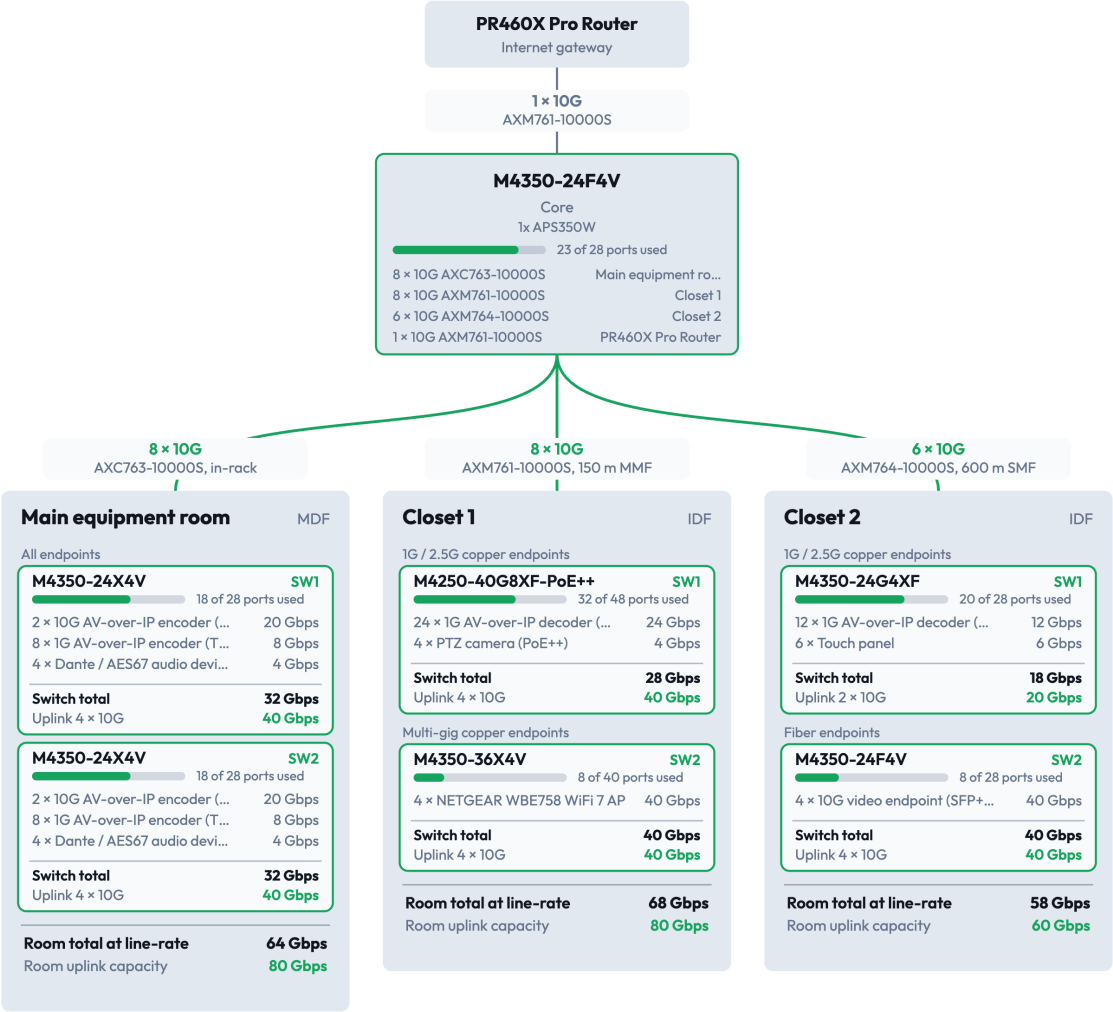
Design summary

Estimate for planning only. This bill of materials is generated automatically as a budgetary estimate. It must be validated by the ProAV Design team before ordering. Validation: proavdesign@netgear.com

ROOMS AND DEVICES

Device	Category	Qty	Link	PoE each
Main equipment room • MDF • 28 devices				
1G AV-over-IP encoder (TX)	Video	16	1G Copper	13 W
10G AV-over-IP encoder (copper)	Video	4	10G Copper	–
Dante / AES67 audio device	Audio	8	1G Copper	13 W
Closet 1 • IDF, 150 m multimode fiber • 32 devices				
1G AV-over-IP decoder (RX)	Video	24	1G Copper	13 W
PTZ camera (PoE++)	Camera	4	1G Copper	60 W
NETGEAR WBE758 WiFi 7 AP	Wireless	4	10G Copper	39 W
Closet 2 • IDF, 600 m single mode fiber • 22 devices				
1G AV-over-IP decoder (RX)	Video	12	1G Copper	13 W
10G video endpoint (SFP+ fiber)	Video	4	10G Fiber	–
Touch panel	Control	6	1G Copper	25 W

Network diagram



Green lines are uplinks, labelled with link count, speed and the optic or cable used. With a redundant core, every switch has its own full-capacity link group to each core, so either core can carry all traffic alone.

Bill of materials

Part number	Description	Qty	Where
SWITCHES			
GSM4248UX-100NAS	M4250-40G8XF-PoE++ switch	1	Closet 1
GSM4328-100NES	M4350-24G4XF switch	1	Closet 2
XSM4328CV-100NES	M4350-24X4V switch	2	Main equipment room
XSM4328FV-100NES	M4350-24F4V switch	2	Closet 2, Main equipment room (core)
XSM4340CV-100NES	M4350-36X4V switch	1	Closet 1
POWER			
APS350W-100NES	350W modular PSU	2	Closet 2, Main equipment room (core)
APS600W-200NES	600W modular PSU (v2)	3	Main equipment room, Closet 2
APS600W-300NES	600W modular PSU (v3)	1	Closet 1
OPTICS			
AXM761-10000S	10GBASE-SR SFP+, multimode	22	Closet 2, Closet 1, Main equipment room (core), Main equipment room
AXM764-10000S	10GBASE-LR LITE SFP+, single mode	12	Closet 2, Main equipment room (core)
CABLING			
AXC763-10000S	10G SFP+ DAC 3m, passive copper	8	Main equipment room
WIRELESS			
WBE758-111NAS	NETGEAR WBE758 WiFi 7 AP	4	Closet 1
GATEWAY			
PR460X-111NAS	PR460X Pro Router	1	Main equipment room

Part numbers are for Americas. Pricing and availability are provided by NETGEAR or your distributor after validation.

Power and rack space

Room	Est. draw	Datasheet max	Heat load	Current	Circuits	UPS size	Rack
Main equipment room	741 W	1,972 W	2,530 BTU/hr	6.7 A at 110 V	1 × 20 A	824 VA+	3.0 U
Closet 1	1,017 W	5,594 W	3,471 BTU/hr	9.2 A at 110 V	1 × 20 A	1,130 VA+	3.0 U
Closet 2	539 W	991 W	1,840 BTU/hr	4.9 A at 110 V	1 × 20 A	599 VA+	2.0 U
Main equipment room (core)	119 W	119 W	407 BTU/hr	1.1 A at 110 V	1 × 20 A	133 VA+	1.0 U

POWER SUPPLIES PER SWITCH

Room	Device	Qty	PoE load each	Power supplies	PoE available (1 PSU failed)	Headroom	Est. draw each
Main equipment room	M4350-24X4V	2	156 W	1x APS600Wv2	396 W	61%	293 W

Room	Device	Qty	PoE load each	Power supplies	PoE available (1 PSU failed)	Headroom	Est. draw each
Closet 1	M4250-40G8XF-PoE++	1	552 W	2 of 3 PSUs powered	720 W	23%	696 W
Closet 1	M4350-36X4V	1	156 W	1x APS600Wv3	280 W	44%	321 W
Closet 2	M4350-24G4XF	1	306 W	1x APS600Wv2	468 W	35%	420 W
Closet 2	M4350-24F4V	1	–	1x APS350W	–	–	119 W
Main equipment room (core)	M4350-24F4V	1	–	1x APS350W	–	–	119 W
Main equipment room	PR460X Pro Router	1	–	Internal PSU	–	–	156 W

Estimated draw = datasheet maximum without PoE + PoE load ÷ 0.9 (PSU efficiency). Circuits assume 80% continuous load. Feed each power supply from a separate circuit or UPS.

Checks

NOTE	PR460X Pro Router does not take DAC/AOC cables, so it connects to the core with AXM761-10000S optics on both ends and a multimode LC patch cord.
NOTE	Devices above 30W need Ultra90 PoE++ (802.3bt) ports; only those switches were considered for them.
NOTE	Switches interconnect with LAG uplinks and IGMP Plus (no stacking), which keeps AVB and PTP available.
NOTE	Uplink LAGs use 1, 2, 4 or 8 links (sizes from Tool_Settings: 1, 2, 4, 8), so LAG hashing spreads traffic evenly.
NOTE	Redundant power: PoE is sized to the budget that remains if one power supply fails.

UPLINK PLAN

Room	Switch	Units	Uplinks each	Speed	Run	Optic / cable
Main equipment room	M4350-24X4V	2	4	10G	in-rack	AXC763-10000S
Closet 1	M4250-40G8XF-PoE++	1	4	10G	MMF, 150 m	AXM761-10000S
Closet 1	M4350-36X4V	1	4	10G	MMF, 150 m	AXM761-10000S
Closet 2	M4350-24G4XF	1	2	10G	SMF, 600 m	AXM764-10000S
Closet 2	M4350-24F4V	1	4	10G	SMF, 600 m	AXM764-10000S